
ME 354: Science & Technology of Thin Films

Course Code	ME 354
Course Name	Science & Technology of Thin Films
L-T-P-C	3-0-0-3
Prerequisites	Nil
Approval	5th Senate (Oct 2013)

Course Description

Solid-state devices form the basis of integrated circuits, which have a variety of electronic, optoelectronic, and magnetic applications. The research in this field is concerned with design, fabrication and characterization of novel materials and devices with sub-micron feature sizes. Their potential applications include very high-speed devices, optical sources and detectors, optoelectronic components and all-optical devices. The design and fabrication of devices and integrated circuits are inextricably related to device physics, solid-state materials, and sophisticated processing techniques.

This course aims to provide an introduction to the science and technology of thin films, with special emphasis on methods to produce thin films and relationships between growth conditions and thin film properties.

Course Contents

Topics include (1) various methods of thin film production, such as evaporation, sputtering and chemical vapour deposition, (2) nucleation and growth processes, (3) dimensional, chemical, and structural characterization of thin films and (4) properties and applications.